

Introduction to Neural and Cognitive Modelling

Assignment 2

Deadline: 17th September, 11:59 PM

General Instructions

1. Implement all simulations in using a Jupyter Notebook for clarity and reproducibility.
2. Label all plot axes clearly and include appropriate units.
3. Unless otherwise specified, use the Hodgkin–Huxley implementation provided in the `neurodynex3.hodgkin_huxley.HH` module.
4. Academic honesty is mandatory. Any plagiarism will result in a score of 0.
5. AI Policy: You are NOT allowed to use AI to generate code, answer assignment questions or write reports. You may use it to understand concepts or as a proofreader to check grammar, punctuation etc.
6. Start early. The deadline is strict, and no extensions will be granted.

Software Setup

If NeuroDynEx is not already installed, install it using:

```
pip install --upgrade neurodynex3
```

If using Google Colab, use:

```
!pip install --upgrade neurodynex3
```

The package can then be imported using, for example:

```
from neurodynex3.hodgkin_huxley import HH
from neurodynex3.tools import input_factory
```

Spike Detection Convention

For quantitative comparisons in this assignment, use a consistent definition of an action potential.

You may count an upward crossing of

$$V = 20 \text{ mV}$$

as one spike. If you use a different reasonable spike-detection criterion, clearly state it and use it consistently throughout your analysis.

Important Note on Voltage Convention

The NeuroDynEx Hodgkin–Huxley implementation follows the classical Hodgkin–Huxley voltage convention, in which the normal resting membrane potential is approximately

$$V = 0 \text{ mV}.$$

Thus, these voltages are measured relative to the resting reference and should not be interpreted directly as modern absolute membrane potentials such as approximately -65 mV .

Simulation and Stimulus Conventions

Unless otherwise specified, all current stimuli in this assignment follow the convention

$$t_{\text{start}} \leq t < t_{\text{end}}$$

That is, the current is applied beginning at t_{start} and is switched off at t_{end} .

Use the following simulation time steps:

- For parameter or current-amplitude sweeps used to locate a firing threshold, use

$$\Delta t = 0.05 \text{ ms}.$$

- For all voltage and gating-variable traces presented in your final report, use

$$\Delta t = 0.01 \text{ ms}.$$

Thus, when a threshold is determined using a sweep, first identify the threshold using $\Delta t = 0.05 \text{ ms}$ and then rerun the relevant representative conditions using $\Delta t = 0.01 \text{ ms}$ for the plots included in your report.

Part A: The Classical Hodgkin–Huxley Model (25 marks)

The classical Hodgkin–Huxley membrane equation is

$$C_m \frac{dV}{dt} = -(I_{\text{Na}} + I_{\text{K}} + I_L) + I_{\text{ext}},$$

where

$$I_{\text{Na}} = \bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}}),$$

$$I_{\text{K}} = \bar{g}_{\text{K}} n^4 (V - E_{\text{K}}),$$

and

$$I_L = g_L(V - E_L).$$

The gating variables have the following roles:

- m : sodium-channel activation,
- h : sodium-channel inactivation,
- n : potassium-channel activation.

Q1: Dynamics of an Action Potential

(10 marks)

Use the standard NeuroDynEx Hodgkin–Huxley neuron.

Apply a depolarizing current pulse with amplitude

$$I_{\text{ext}} = 7.2 \mu\text{A}$$

during

$$10 \text{ ms} \leq t < 12 \text{ ms},$$

and simulate the neuron for at least 50 ms.

Use a simulation time step of

$$\Delta t = 0.01 \text{ ms}.$$

Plot, as functions of time:

- the injected current $I_{\text{ext}}(t)$,
- the membrane voltage $V(t)$,
- the gating variables $m(t)$, $h(t)$, and $n(t)$.

Clearly label all axes and indicate appropriate units.

Q2: Effect of Sodium Conductance on Neuronal Excitability

(15 marks)

The maximum sodium conductance $\bar{g}_{\text{Na}}$ determines the strength of the inward sodium current when sodium channels are open.

For this question, compare

Standard HH: $\bar{g}_{\text{Na}}$

with

Modified HH: $1.4 \bar{g}_{\text{Na}}$.

Obtaining and Modifying the Hodgkin–Huxley Source Code

For this question, you will modify the Hodgkin–Huxley implementation provided in the `neurodynex3` package.

The source code for the model is available in the official NeuroDynEx GitHub repository:

https://github.com/EPFL-LCN/neurondynamics-exercises/blob/master/neurodynex3/hodgkin_huxley/HH.py

The relevant function is

`simulate_HH_neuron()`

which contains the Brian2 implementation of the Hodgkin–Huxley neuron, including the membrane equation, ionic conductances, reversal potentials, and gating-variable dynamics.

Follow these steps:

1. Open the source-code link above.
2. Locate the function `simulate_HH_neuron()`.
3. Copy the complete function into your Jupyter Notebook.
4. Rename your copied function, for example:

```
simulate_modified_HH_neuron()
```

so that the original NeuroDynEx function remains unchanged.

5. Modify the maximum sodium conductance inside your copied function according to

$$\bar{g}_{\text{Na}} \rightarrow 1.4 \bar{g}_{\text{Na}}.$$

6. Use the original NeuroDynEx function for the standard HH neuron and your copied function for the modified neuron.

Important: Do not modify the installed `neurodynex3` package files directly. Make a copy of the function in your own notebook and modify that copy.

Experiment

Apply a constant depolarizing step current during

$$50 \text{ ms} \leq t < 350 \text{ ms},$$

and simulate the neuron for a total of 400 ms.

Define **repetitive firing** as at least three action potentials occurring during the current step.

For the threshold search, test current amplitudes over the range

$$0 \leq I_{\text{ext}} \leq 15 \mu\text{A}$$

in increments of

$$0.1 \mu\text{A}.$$

Use a simulation time step of

$$\Delta t = 0.05 \text{ ms}$$

for the threshold search.

- (a) Determine the minimum step-current amplitude that produces repetitive firing in the standard HH neuron.
- (b) Repeat the analysis after increasing the sodium conductance to

$$\bar{g}_{\text{Na,new}} = 1.4 \bar{g}_{\text{Na,standard}}.$$

- (c) Report the two threshold currents and provide representative voltage traces for both neurons near their respective repetitive-firing thresholds.

Rerun the conditions used for the reported voltage traces using

$$\Delta t = 0.01 \text{ ms}.$$

Part B: Extending the Hodgkin–Huxley Model with a New Ion Channel (25 marks)

The standard Hodgkin–Huxley neuron contains voltage-gated sodium and potassium currents together with a leak current. Real neurons express many additional voltage-dependent ion channels.

The Hodgkin–Huxley framework can be extended by adding an additional ionic current:

$$C_m \frac{dV}{dt} = -(I_{\text{Na}} + I_{\text{K}} + I_L + I_X) + I_{\text{ext}},$$

where I_X represents the current produced by an additional ion channel.

Each group will investigate one specific Hodgkin–Huxley-type ion-channel model from [Channelpedia](#).

Assigned Channels

Student Name	Group	Channel	Channelpedia Model	Assigned $\bar{g}_X$
Ashish Beniwal Soham Chatterjee Parth Dhodapkar Manali Gupta	1	HCN1	Model ID 9	2 mS
Arjun Kalidindi Kavish Kapoor Athmeeya Kashyap Joshua Koilpillai	2	Kv1.1	Model ID 18	0.25 mS
Laksh Mittal Princy Patel Sahishnu Pawankumar Inesh Roy	3	Kv3.1_S	Model ID 27	40 mS
Kashik S Chandrani Saha Nisarg Suthar SaiVeeksha Tavva	4	Cav3.1	Model ID 41	20 mS

You should work on the assignment individually, NOT as a group.

(The assigned conductance values above are values chosen for this assignment. They should not be interpreted as measured channel densities for the squid giant axon)

The relevant Channelpedia pages are:

- **HCN1:** <https://channelpedia.epfl.ch/wikipages/61>
- **Kv1.1:** <https://channelpedia.epfl.ch/ionchannels/1>
- **Kv3.1:** <https://channelpedia.epfl.ch/ionchannels/11>
- **Cav3.1:** <https://channelpedia.epfl.ch/wikipages/85>

Very Important: Voltage Convention

The Channelpedia models and the NeuroDynEx implementation use different voltage conventions.

The Channelpedia gating equations use absolute membrane voltage, whereas the NeuroDynEx implementation follows the classical Hodgkin–Huxley convention in which rest is approximately 0 mV.

Therefore, when evaluating the Channelpedia gating equations, convert the NeuroDynEx voltage according to

$$V_{\text{CP}} = V_{\text{HH}} - 65 \text{ mV}$$

where V_{CP} is the voltage that should be inserted into the Channelpedia expressions for $x_{\infty}(V)$ and $\tau_x(V)$.

Similarly, convert the reversal potential supplied by Channelpedia according to

$$E_{X,\text{HH}} = E_{X,\text{CP}} + 65 \text{ mV}$$

before using it in the NeuroDynEx membrane equation.

For example, HCN1 Model ID 9 specifies

$$E_{\text{HCN},\text{CP}} = -45 \text{ mV}.$$

Therefore, in the NeuroDynEx voltage convention,

$$E_{\text{HCN},\text{HH}} = -45 + 65 = 20 \text{ mV}.$$

Baseline Hodgkin–Huxley Model for Part B

All simulations in Part B must use the **original standard Hodgkin–Huxley neuron** as the baseline model.

In particular, use

$$\bar{g}_{\text{Na}} = 120 \text{ mS}$$

for all Part B simulations.

The modification

$$\bar{g}_{\text{Na}} \rightarrow 1.4 \bar{g}_{\text{Na}}$$

is used **only in Q2** and must not be carried over into Part B.

The modified neurons in Part B therefore differ from the standard HH neuron only by the addition of the assigned ion channel and its associated gating variable(s).

Implementation Instructions for Part B

For Part B, begin with a fresh copy of the **standard** NeuroDynEx `simulate_HH_neuron()` implementation. Do not begin with the $1.4 \times \bar{g}_{\text{Na}}$ function created for Q2.

Create a new function for the neuron containing your assigned channel. For example,

```
simulate_HH_with_new_channel()
```

Modify this copied implementation by:

1. adding the new ionic current I_X to the membrane equation;
2. adding the required gating-variable differential equation(s);
3. using the steady-state functions and time constants from the **specific Channelpedia Model ID assigned to your group**;
4. using the assigned maximum conductance $\bar{g}_X$ given in the table above.

Do **not** substitute another model listed for the same channel.

You do not need to compile or directly use the Channelpedia `.mod` file. Implement the Hodgkin–Huxley gating equations given on the Channelpedia page directly in your copied NeuroDynEx model.

Do not apply additional temperature or Q_{10} corrections.

Initialize each additional gating variable x at its steady-state value corresponding to the initial absolute membrane voltage:

$$x(0) = x_{\infty}(-65 \text{ mV}).$$

Q3: Characterizing the Assigned Ion Channel (10 marks)

Before adding the channel to the complete neuron, investigate the properties of the channel itself.

(a) From the assigned Channelpedia model, report:

- the ion(s) or ionic current carried by the channel,
- its reversal potential,
- its activation and/or inactivation variables,
- the power of each gating variable.

Write the ionic-current equation for your assigned channel in the form

$$I_X = \bar{g}_X m^p h^q (V - E_X),$$

or the appropriate simplified form if your channel has only one gating variable.

(b) Write the differential equation governing each gating variable in the form

$$\frac{dx}{dt} = \frac{x_{\infty}(V) - x}{\tau_x(V)}.$$

Identify the expressions for $x_{\infty}(V)$ and $\tau_x(V)$ from your assigned Channelpedia model.

(c) Over the voltage range

$$-120 \text{ mV} \leq V_{\text{CP}} \leq 60 \text{ mV},$$

plot:

- the steady-state activation curve(s),
- the steady-state inactivation curve, if present,
- the corresponding gating time constant(s).

You may use separate plots when the quantities have substantially different scales.

(d) Based only on the gating curves and time constants, predict:

At what membrane potentials should your channel become most important, and how do you expect adding it to alter the behaviour of the neuron?

Support your prediction using the voltage dependence and time scale of the channel.

Q4: Functional Effect of the Assigned Ion Channel (15 marks)

Add the assigned ionic current I_X and its gating-variable differential equation(s) to your copied Hodgkin–Huxley implementation.

For the experiment assigned to your group, compare

$$\boxed{\text{Standard HH}} \quad \text{with} \quad \boxed{\text{HH} + \text{assigned channel}}.$$

Perform **only the experiment assigned to your group**.

Group 1: HCN1

Apply a hyperpolarizing step current

$$I_{\text{ext}} = -10 \mu\text{A}$$

during

$$50 \text{ ms} \leq t < 250 \text{ ms},$$

and simulate for a total of 300 ms.

Use

$$\Delta t = 0.01 \text{ ms.}$$

Plot:

- the injected current $I_{\text{ext}}(t)$,
- $V(t)$ for the standard HH neuron and the HH+HCN1 neuron,
- the HCN1 activation variable.

Measure the **voltage-sag amplitude**, defined as

$$V_{\text{sag}} = V_{\text{late}} - V_{\text{min}},$$

where V_{min} is the minimum membrane voltage reached after the beginning of the hyperpolarizing step and V_{late} is the approximately steady voltage near the end of the step.

Group 2: Kv1.1

Apply depolarizing step currents during

$$50 \text{ ms} \leq t < 350 \text{ ms},$$

and simulate for a total of 400 ms.

For the threshold search, test current amplitudes over

$$0 \leq I_{\text{ext}} \leq 15 \mu\text{A}$$

in increments of

$$0.1 \mu\text{A}.$$

Use

$$\Delta t = 0.05 \text{ ms}$$

for the threshold search.

Determine the minimum current producing at least three action potentials during the current step for:

1. the standard HH neuron,
2. the HH neuron containing Kv1.1.

Report the two threshold currents.

Plot representative voltage traces near the respective thresholds together with the Kv1.1 activation and inactivation variables.

Rerun the conditions used for the reported traces using

$$\Delta t = 0.01 \text{ ms.}$$

Group 3: Kv3.1_S

Apply

$$I_{\text{ext}} = 10 \text{ } \mu\text{A}$$

during

$$50 \text{ ms} \leq t < 350 \text{ ms,}$$

and simulate for 400 ms using

$$\Delta t = 0.01 \text{ ms.}$$

Plot:

- $V(t)$ for the standard HH neuron and the HH+Kv3.1_S neuron,
- the Kv3.1_S activation variable.

For the **first** action potential during the current step, measure the action-potential width at half-amplitude.

Define the half-amplitude voltage relative to the membrane voltage immediately before the spike:

$$V_{1/2} = V_{\text{baseline}} + \frac{V_{\text{peak}} - V_{\text{baseline}}}{2}.$$

Measure the interval between the upward and downward crossings of $V_{1/2}$ and compare the resulting spike width between the standard and modified neurons.

Important: Use **Kv3.1_S**, **Model ID 27**. Do not use Model ID 26.

Group 4: Cav3.1

Apply a hyperpolarizing step current

$$I_{\text{ext}} = -2 \text{ } \mu\text{A}$$

during

$$50 \text{ ms} \leq t < 250 \text{ ms},$$

then remove the current and simulate the neuron until at least

$$t = 330 \text{ ms}.$$

Use

$$\Delta t = 0.01 \text{ ms}.$$

Plot:

- the injected current $I_{\text{ext}}(t)$,
- $V(t)$ for the standard HH neuron and the HH+Cav3.1 neuron,
- the Cav3.1 activation and inactivation variables.

Determine whether an action potential occurs following release from hyperpolarization in either model.

If a rebound action potential occurs, report its latency relative to the removal of the hyperpolarizing current at

$$t = 250 \text{ ms}.$$

Final Interpretation — All Groups

Answer the following in approximately **150–250 words**:

How did the addition of your assigned ion channel change the behaviour of the Hodgkin–Huxley neuron? Explain the effect using the channel’s voltage dependence, gating time scale, ionic reversal potential, and the evolution of its gating variable(s). Was the prediction you made in Q3 supported by the simulation?

Your answer should make a connection between

$$\boxed{\text{Channel kinetics}} \longrightarrow \boxed{\text{Ionic current}} \longrightarrow \boxed{\text{Neuronal behaviour}}.$$

Submission Checklist

Your submission should contain:

- A Jupyter Notebook (`.ipynb`) containing all simulation and analysis code.
- Clearly labelled plots with appropriate units.

- For **Part A**:
 - the current, voltage, and m , h , n traces from Q1;
 - repetitive-firing thresholds for the standard and modified HH neurons;
 - representative near-threshold voltage traces.
- For **Part B**:
 - the assigned channel equation and gating equations;
 - steady-state gating curves and time constants;
 - your prediction based on the channel kinetics;
 - the standard-HH versus modified-HH comparison;
 - the quantitative measurement assigned to your group;
 - the new channel's gating-variable traces;
 - the final mechanistic interpretation.

Grading Scheme

Evaluation will be conducted individually based on code and output review, and viva. You should be prepared to walk through your code, interpret your plots, and explain the underlying Hodgkin–Huxley mechanisms.

Component	Marks
Part A: Classical Hodgkin–Huxley Model	25
Part B: Additional Ion Channel	25
Viva	50
Total	100

Good luck!